



CAPSTONE PROJECT REGISTER

Class: IAP491 Duration time: from 18/11/2025 To 25/04/2026

(*) Profession: Information Assurance Specialty: <ES> ☐ <IS> ☐ <JS> ☐

(*) Kinds of person make registers: Lecturer ☒ Students ☐

1. Register information for supervisor (if have)

	Full name	Phone	E-Mail	Title
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Supervisor 2				

2. Register information for students (if have)

	Full name	Student code	Phone	E-mail	Role in Group
Student 1	Hồ Lê Bình	SE183564	0325683908	binhlse183564@fpt.edu.vn	Leader
Student 2	Nguyễn Hoàng Trí	SE183413	0965916742	trinhse183413@fpt.edu.vn	Member
Student 3	Phạm Tuấn Anh	SE183403	0326864535	anhptse183403@fpt.edu.vn	Member
Student 4	Trịnh Nguyễn Yến Vy	SE183776	0385800046	vytnyse183776@fpt.edu.vn	Member

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

English: Adaptive AI Defense Agent (Reinforcement Learning -based)

Vietnamese: Nghiên cứu và phát triển AI Agent có khả năng tự phòng thủ thích ứng trong môi trường mô phỏng dựa trên kỹ thuật Reinforcement Learning.

Abbreviation: RL – Defense - Agent

(*) 3.2. Main proposal content (including result and product)

a) Theory and practice (document):

Introduction Overview

Cyber threats are evolving rapidly with increasing complexity and automation. Traditional defense mechanisms, such as rule-based firewalls or static intrusion detection systems (IDS), often fail to detect novel attacks (zero-day) or adapt to dynamic attack strategies in real-time.

To address this challenge, Artificial Intelligence (AI), specifically Reinforcement Learning (RL), offers a promising approach. Unlike supervised learning which requires labeled datasets, RL agents learn optimal defense strategies through interaction with the environment (trial and error).

This project focuses on building an Adaptive AI Defense Agent. The agent will be trained in a simulated network environment (using OpenAI Gymnasium framework). It will learn to observe the network state, detect malicious activities, and autonomously execute defense actions to maximize system security while maintaining service availability.

Objective of the project:

- Research the application of Reinforcement Learning (RL) in Cyber Security.
- Build a simulated network environment for training defensive agents (using Gymnasium).
- Develop and train an RL Agent capable of automatically detecting and mitigating attacks.
- Evaluate the agent's performance against baseline attackers in the simulation.

b) Program:

An AI-based software agent capable of autonomous decision-making for network defense within a simulation environment.

Functional Requirement

(1) Simulation Environment Module:

- Real-world Attack Simulation: Utilizes actual security tools (Kali Linux suite) to generate multi-vector attacks aligned with MITRE ATT&CK, including:
 - DoS/DDoS: TCP SYN Flood, UDP Flood, HTTP Flood (via [hping3](#), [slowloris](#)).
 - Reconnaissance: Port Scanning, Service Version Detection (via [nmap](#)).
 - Web/Auth Attacks: Brute Force, SQL Injection payloads (via [hydra](#), custom scripts).
- Advanced Enforcement Mechanism: Executes granular defensive actions via Linux Kernel (Iptables/TC):
 - Blocking: Drop packets from malicious IPs.
 - Rate Limiting: Throttle bandwidth for suspicious traffic to mitigate impact.
 - Deception (Honeypot Redirection): Redirect malicious flows to isolated Honeypot containers using NAT.

(2) Agent Training Module:

- Extended Action Space: The Agent selects from discrete actions: [0: Allow, 1: Rate Limit, 2: Redirect to Honeypot, 3: Block].

- Deep Reinforcement Learning: Implements PPO/DQN algorithms to learn optimal defense policies, balancing between security and service availability.
- Feature Extraction: Converts raw captured packets into numerical feature vectors for the Neural Network.

(3) Evaluation and Visualization (SIEM Integrated):

- Centralized Dashboard: Uses Wazuh SIEM to visualize real-time system status.
- Threat Intelligence Display: Shows Attack Source, Attack Type, and AI Countermeasures (e.g., "Blocked IP X", "Redirected IP Y to Honeypot").
- Performance Metrics: Tracks False Positive Rate and Deception Success Rate.

(4) System Management:

- Log Integration: Generates structured JSON logs from the AI module for SIEM ingestion.
- Container Management: Orchestrates the lifecycle of Victim Servers and Honeypot containers.

Tools/Platform

- **Language:** Python
- **RL Framework:** PyTorch, Stable Baselines3
- **Simulation:** Gymnasium, Mininet
- **Network & Security:** Scapy, Iptables, Docker
- **SIEM & Visualization:** Wazuh
- **Attack Generation:** Kali Linux Tools
- **IDE:** VSCode

Hardware

- **Operating System:** Ubuntu Linux 20.04/22.04 LTS
- **Processor:** Intel Core i5 or AMD Ryzen 5 or higher
- **GPU:** NVIDIA RTX Series
- **Memory:** 24 GB+ RAM

c) Other products:

4. Other comment (propose all relative thing if have)

Proposed Tasks for students:

Student 1: Infrastructure & Attack Simulation Specialist

- Network Construction: Construct the virtualized network topology using Mininet, comprising Production Servers, Client Nodes, and isolated Honeypot Containers (Docker).
- Attack Automation: Develop Python scripts to automate standard security tools (Nmap, Hping3, Hydra) for generating realistic multi-vector attack datasets (DoS, Scanning, Brute-force).

- Enforcement Implementation: Implement Python modules to execute low-level Linux kernel commands (`iptables` for Blocking/NAT Redirection, `tc` for Rate Limiting) triggered by the AI Agent.

Student 2: AI & Reinforcement Learning Core Researcher

- Model Architecture: Design the Deep Neural Network (DNN) architecture and implement Reinforcement Learning algorithms (PPO or DQN).
- Reward Engineering: Design and fine-tune a complex Reward Function to guide the Agent (e.g., high positive rewards for successful Honeypot redirection, severe negative penalties for blocking legitimate users).
- Agent Training: Conduct model training and hyperparameter tuning using the feature datasets provided by Student 4 to ensure convergence and optimal policy learning.

Student 3: SIEM Engineer (Wazuh Specialist)

- SIEM Deployment: Deploy and configure the Wazuh Manager (Server) using Docker and install Wazuh Agents on network nodes.
- Log Integration: Configure Custom Decoders and Rules in Wazuh to parse and index the specific JSON logs generated by the AI module.
- Dashboard Design: Create a comprehensive Security Dashboard in the Wazuh interface to visualize real-time attack vectors, AI decisions, and system health status for the final demonstration.

Student 4: System Integrator & Feature Engineer

- Sniffer & Parser Development: Develop the Sniffer Module using `Scapy` to capture real-time network packets and perform Feature Extraction (converting raw packets into numerical vectors) for the AI input.
- Logging Module: Develop a structured logging system to record AI actions (Block/Redirect/Allow) into standard JSON format for SIEM ingestion.
- System Integration: Orchestrate the connection between the Sniffer, AI Model, Enforcement Scripts (Student 1), and SIEM (Student 3) to ensure a seamless real-time pipeline.

References:

- [1] Huang, H., Cheng, L., Zhou, X., & Peng, D. (2024). *Reinforcement learning-based zero-day vulnerability detection model*. *Sensors*, 24(20).
- [2] Banik, S., & Dandyala, S. S. M. (2024). *Reinforcement learning for adaptive cybersecurity: An overview*. *International Journal of Machine Learning Research, Computing, and Artificial Intelligence*, 4(2).
- [3] Gopireddy, R. R. (2023). *Reinforcement learning for cyber defense: Adaptive and autonomous security systems*. *European Journal of Advances in Engineering and Technology*, 10(10), 102–105.

[4] Romano, G., & Klein, T. (2024). *Reinforcement learning for adaptive cybersecurity*.

Tp. Hồ Chí Minh, date 18/11/2025

Supervisor (If have)

(Sign and full name)

On behalf of Registers

(Sign and full name)

Yến Vy

Trịnh Nguyễn Yến Vy